

LAB 2

TITLE: Creating table in MySQL database and performing the given queries.

OBJECTIVE: Create a table as the following schema and run the queries as given below.

student(stud_id, full_name, address, contact, age, gender, major_subject)

Queries:

- a. Select all records.
- b. Select records of all the males.
- c. Select name and address of all the female whose age is between 15 and 25.
- d. Display the stud_id and name of all the students in alphabetical order.
- e. Display the records of students sorted by age in ascending order.
- f. Display the total number of students.
- g. Display the number of male and female students.
- h. Display the average age of all the male and female students.
- i. Display the distinct major subjects.
- j. Display the number of major subjects taken by students.
- k. Display the name of youngest student.
- l. Name of all the students whose name starts from 's' and are above 19 years.

THEORY:

MySQL: MySQL is an open-source relational database management system (RDBMS). It is widely used for storing, managing, and retrieving data in a structured format, typically in tables composed of rows and columns.

SQL: SQL, which stands for Structured Query Language, is a standardized programming language designed for managing and manipulating data stored in relational databases. It serves as the primary means of communication with these databases, allowing users to perform various operations on the data. Languages of SQL are:

1. **Data Definition Language (DDL):** DDL commands are used to define and manage the structure of database objects.
 - CREATE: Used to create database objects like tables, views, or indexes.
 - ALTER: Used to modify the structure of existing database objects.
 - DROP: Used to delete database objects.
 - TRUNCATE: Used to remove all records from a table while retaining its structure.
2. **Data Manipulation Language (DML):** DML commands are used to manage the data within database objects.
 - INSERT: Used to add new records into a table.
 - UPDATE: Used to modify existing records in a table.
 - DELETE: Used to remove records from a table.
3. **Data Query Language (DQL):** DQL commands are primarily used for retrieving data from the database.
 - SELECT: Used to query and retrieve data from one or more tables.
4. **Data Control Language (DCL):** DCL commands are used to control access and permissions within the database.
 - GRANT: Used to give users or roles specific privileges.
 - REVOKE: Used to remove previously granted privileges.
5. **Transaction Control Language (TCL):** TCL commands are used to manage transactions, ensuring data consistency and integrity.
 - COMMIT: Used to save all changes made during a transaction.
 - ROLLBACK: Used to undo changes made during a transaction.
 - SAVEPOINT: Used to set a point within a transaction to which you can later roll back.

```

C:\XAMPP for Windows - mysql -u root -p
MariaDB [school]> create table student(
-> stud_id int auto_increment primary key,
-> full_name varchar(30) not null,
-> address varchar(50) not null,
-> contact varchar(50) not null,
-> age int not null,
-> gender enum('male','female'),
-> major_subject varchar(50) not null
-> );
Query OK, 0 rows affected (0.018 sec)

```

Creating Student Table

OUTPUTS:

a) Select all records

```
MariaDB [school]> SELECT * FROM student;
```

stud_id	full_name	address	contact	age	gender	major_subject
1	Aarav Sharma	Kathmandu, Nepal	9812345670	20	male	Computer Science
2	Priya Koirala	Pokhara, Nepal	9801234561	18	female	Information Technology
3	Suman Thapa	Lalitpur, Nepal	9812345672	21	male	Civil Engineering
4	Anisha Shrestha	Biratnagar, Nepal	9801234563	17	female	Business Studies
5	Ramesh Adhikari	Chitwan, Nepal	9812345674	19	male	Computer Science
6	Nisha KC	Bhaktapur, Nepal	9801234565	22	female	Economics
7	Dipesh Basnet	Dharan, Nepal	9812345676	24	male	Mechanical Engineering
8	Sneha Maharjan	Hetauda, Nepal	9801234567	15	female	Computer Engineering
9	Prakash Gurung	Butwal, Nepal	9812345678	26	male	Electrical Engineering
10	Bimala Rai	Janakpur, Nepal	9801234569	16	female	Sociology
11	Roshan Lama	Banepa, Nepal	9812345680	25	male	Software Engineering
12	Kusum Bhandari	Damak, Nepal	9801234571	18	female	Nursing
13	Kiran Shahi	Nepalgunj, Nepal	9812345682	20	male	Physics
14	Ritika Poudel	Gorkha, Nepal	9801234573	17	female	Chemistry

b) Select records of all the males

```
MariaDB [school]> SELECT * FROM student WHERE gender= 'male';
```

stud_id	full_name	address	contact	age	gender	major_subject
1	Aarav Sharma	Kathmandu, Nepal	9812345670	20	male	Computer Science
3	Suman Thapa	Lalitpur, Nepal	9812345672	21	male	Civil Engineering
5	Ramesh Adhikari	Chitwan, Nepal	9812345674	19	male	Computer Science
7	Dipesh Basnet	Dharan, Nepal	9812345676	24	male	Mechanical Engineering
9	Prakash Gurung	Butwal, Nepal	9812345678	26	male	Electrical Engineering
11	Roshan Lama	Banepa, Nepal	9812345680	25	male	Software Engineering
13	Kiran Shahi	Nepalgunj, Nepal	9812345682	20	male	Physics
15	Ashish Magar	Palpa, Nepal	9812345684	19	male	Mathematics
17	Rajendra Oli	Kailali, Nepal	9812345686	23	male	Biology
19	Gopal Pandey	Kavre, Nepal	9812345688	15	male	Economics
21	Dinesh KC	Parsa, Nepal	9812345690	22	male	Political Science
23	Hari Khadka	Rasuwa, Nepal	9812345692	19	male	Civil Engineering
25	Sandeep Jha	Saptari, Nepal	9812345694	26	male	Mathematics
27	Manoj Rimal	Dolakha, Nepal	9812345696	25	male	Biology
29	Keshav Yadav	Siraha, Nepal	9812345698	20	male	Physics
31	Santosh KC	Okhaldhunga, Nepal	9812345700	17	male	Computer Engineering
33	Sunil Adhikari	Tanlejung, Nepal	9812345702	19	male	Software Engineering

c) Select name and address of all the female whose age is between 15 and 25

```
MariaDB [school]> SELECT full_name, address FROM student WHERE gender='female' AND age>15 AND age<25;
```

full_name	address
Priya Koirala	Pokhara, Nepal
Anisha Shrestha	Biratnagar, Nepal
Nisha KC	Bhaktapur, Nepal
Bimala Rai	Janakpur, Nepal
Kusum Bhandari	Damak, Nepal
Ritika Poudel	Gorkha, Nepal
Sandhya Ale	Surkhet, Nepal
Manisha Tamang	Sindhupalchok, Nepal
Sujata Rana	Baglung, Nepal
Asmita Thakuri	Darchula, Nepal
Bhawana Shahi	Dadeldhura, Nepal

QUERIES

- a) Select all records
 - `SELECT * FROM student;`
- b) Select records of all the males.
 - `SELECT * FROM student WHERE gender= 'male';`
- c) Select name and address of all the female whose age is between 15 and 25.
 - `SELECT full_name, address FROM student WHERE gender='female' AND age>15 AND age<25;`
- d) Display the stud_id and name of all the students in alphabetical order.
 - `SELECT stud_id, full_name FROM student ORDER BY full_name ASC;`
- e) Display the records of students sorted by age in ascending order.
 - `SELECT * FROM student ORDER BY age ASC;`
- f) Display the total number of students.
 - `SELECT COUNT(*) AS tot_student FROM student;`
- g) Display the number of male and female students.
 - `SELECT gender, COUNT(*) AS tot_student FROM student GROUP BY gender;`
- h) Display the average age of all the male and female students.
 - `SELECT gender, AVG(age) AS average_age FROM student GROUP BY gender;`
- i) Display the distinct major subjects.
 - `SELECT DISTINCT major_subject FROM student;`
- j) Display the number of major subjects taken by students.
 - `SELECT COUNT(DISTINCT major_subject) AS total_major_subjects FROM student;`
- k) Display the name of youngest student.
 - `SELECT full_name, age FROM student ORDER BY age ASC LIMIT 1;`
- l) Name of all the students whose name starts from 's' and are above 19 years.
 - `SELECT full_name, age FROM student WHERE full_name LIKE 's%' AND age > 19;`

OUTPUTS:

d) Display the stud_id and name of all the students in alphabetical order

```
MariaDB [school]> SELECT stud_id, full_name FROM student ORDER BY full_name ASC;
```

stud_id	full_name
1	Aarav Sharma
4	Anisha Shrestha
44	Anita Tamang
15	Ashish Magar
22	Asmita Thakuri
24	Bhawana Shahi
37	Bikash Magar
10	Bimala Rai
34	Bimala Shrestha
40	Binita Gurung
47	Bipin Karki
42	Deepa Ale
21	Dinesh KC
7	Dipesh Basnet
19	Gopal Pandey
23	Hari Khadka

e) Display the records of students sorted by age in ascending order.

```
MariaDB [school]> SELECT * FROM student ORDER BY age ASC;
```

stud_id	full_name	address	contact	age	gender	major_subject
28	Srijana Sunar	Bajhang, Nepal	9801234587	15	female	Chemistry
8	Sneha Maharjan	Hetauda, Nepal	9801234567	15	female	Computer Engineering
42	Deepa Ale	Bajura, Nepal	9801234601	15	female	Business Studies
35	Kamal Thapa	Ramechhap, Nepal	9812345704	15	male	Political Science
19	Gopal Pandey	Kavre, Nepal	9812345688	15	male	Economics
43	Rajan Poudel	Achham, Nepal	9812345712	16	male	Mechanical Engineering
16	Sandhya Ale	Surkhet, Nepal	9801234575	16	female	English Literature
10	Bimala Rai	Janakpur, Nepal	9801234569	16	female	Sociology
20	Sujata Rana	Baglung, Nepal	9801234579	17	female	Computer Science
31	Santosh KC	Okhaldhunga, Nepal	9812345700	17	male	Computer Engineering
14	Ritika Poudel	Gorkha, Nepal	9801234573	17	female	Chemistry
50	Sita Bista	Salyan, Nepal	9801234609	17	female	Computer Engineering
40	Binita Gurung	Lamjung, Nepal	9801234599	17	female	English Literature
4	Anisha Shrestha	Biratnagar, Nepal	9801234563	17	female	Business Studies
12	Kusum Bhandari	Damak, Nepal	9801234571	18	female	Nursing

f) Display the total number of students.

```
MariaDB [school]> SELECT COUNT(*) AS tot_student FROM student;
```

tot_student
50

g) Display the number of male and female students.

```
MariaDB [school]> SELECT gender, COUNT(*) AS tot_student FROM student GROUP BY gender;
```

gender	tot_student
male	25
female	25

h) Display the average age of all the male and female students.

```
MariaDB [school]> SELECT gender, AVG(age) AS average_age FROM student GROUP BY gender;
```

gender	average_age
male	20.8400
female	18.9200

i) Display the distinct major subjects.

```
MariaDB [school]> SELECT DISTINCT major_subject FROM student;
```

major_subject
Computer Science
Information Technology
Civil Engineering
Business Studies
Economics
Mechanical Engineering
Computer Engineering
Electrical Engineering
Sociology

j) Display the number of major subjects taken by students.

```
MariaDB [school]> SELECT COUNT(DISTINCT major_subject) AS total_major_subjects FROM student;
```

total_major_subjects
18

k) Display the name of youngest student.

```
MariaDB [school]> SELECT full_name, age FROM student ORDER BY age ASC LIMIT 1;
```

full_name	age
Sneha Maharjan	15

l) Name of all the students whose name starts from 's' and are above 19 years.

```
MariaDB [school]> SELECT full_name, age FROM student WHERE full_name LIKE 's%' AND age > 19;
```

full_name	age
Suman Thapa	21
Sandeep Jha	26
Sabina Lama	22
Sudip Rijal	25
Sushant BK	22

CONCLUSION:

The lab work gives the hand on experience in SQL commands and can successfully display the tables in terminal. The lab work is done for different SELECT commands to display tables and the valuable data from the student table.